

EXPERIMENT-3

Implement the Decision Tree (ID3) algorithm and classify a new sample.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
from google.colab import files

print("="*70)
print("      DECISION TREE (ID3)")
print("="*70)

uploaded = files.upload()

data = pd.read_csv("PlayTennis.csv")

print("\nDataset:\n")
print(data)

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

encoder = LabelEncoder()

for col in X.columns:
    X[col] = encoder.fit_transform(X[col])
```

```

y = encoder.fit_transform(y)

model = DecisionTreeClassifier(criterion='entropy')

model.fit(X, y)

print("\nDecision Tree Successfully Generated")

print("\nDecision Rules:\n")

print(export_text(model, feature_names=list(X.columns)))

sample = [[2,2,0,1]]

prediction = model.predict(sample)

print("\nTesting New Sample")
print(sample)
print("\nPrediction")

if prediction[0]==1:
    print("Play Tennis = Yes")
else:
    print("Play Tennis = No")

```

Sample Input:

```

... =====
                DECISION TREE (ID3)
=====
Choose Files PlayTennis.csv
PlayTennis.csv(text/csv) - 434 bytes, last modified: 8/5/2026 - 100% done
Saving PlayTennis.csv to PlayTennis.csv

```

Sample Output:

Dataset:

...

	Outlook	Temperature	Humidity	Wind	PlayTennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

Decision Tree Successfully Generated

Decision Rules:

```
|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook > 0.50
|   |--- Humidity <= 0.50
|       |--- Outlook <= 1.50
|           |--- Wind <= 0.50
|               |--- class: 0
|               |--- Wind > 0.50
|                   |--- class: 1
|           |--- Outlook > 1.50
|               |--- class: 0
|   |--- Humidity > 0.50
|       |--- Wind <= 0.50
|           |--- Temperature <= 1.00
|               |--- class: 0
|               |--- Temperature > 1.00
|                   |--- class: 1
|       |--- Wind > 0.50
|           |--- class: 1
```

Testing New Sample

[[2, 2, 0, 1]]

Prediction

Play Tennis = No